

Visualizing Global GDP

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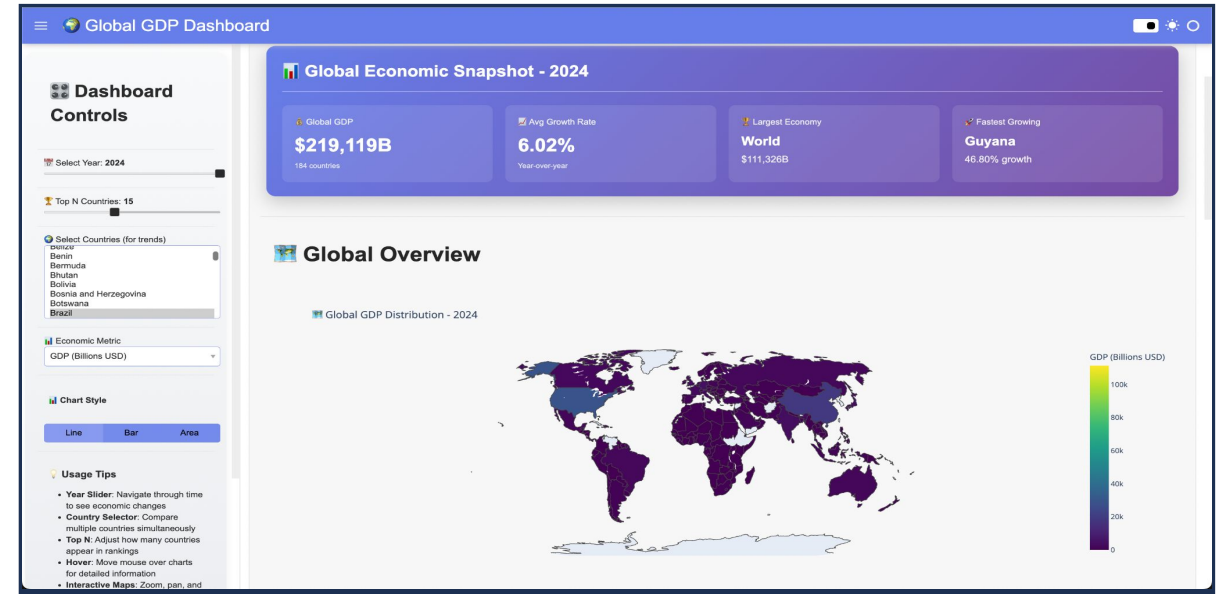
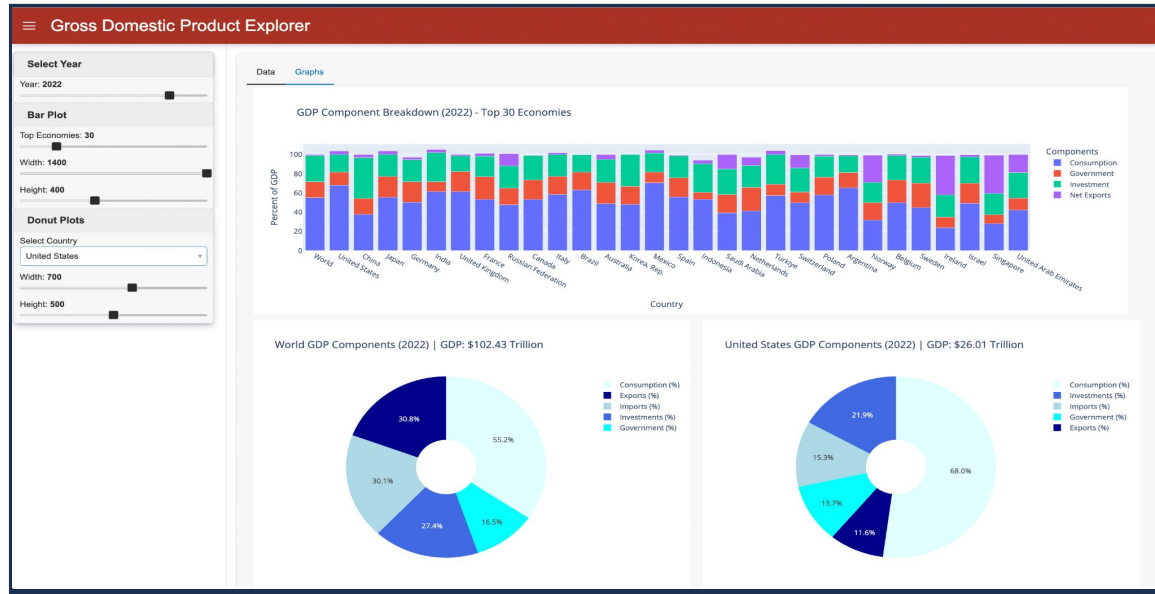


Figure 1. Custom-coded Interactive Dashboard

This dashboard visualizes global GDP component trends with a stacked bar chart and donut charts. This dashboard offers insights into the distribution of GDP components in the top economies, the global breakdown, and a country of choice, all for a specified year from 2014-2024. This data is significant because it allows insights into the breakdown of GDP components of a top economy compared to others. For example, the United States' GDP is made up of 68% consumption, around 40% greater than Ireland's 24%. This reveals fundamentally different economic structures even though both countries fall within the top 30 economies in the world. The donut plots offer the user insights into a country of their choosing compared to the global GDP component makeup, enabling direct comparison between individual economies and worldwide averages. My dashboard allows for the graphs to be adjusted by width and height to allow easy access to detailed information without overwhelming the viewer.

Figure 2. Claude AI-generated Dashboard

Claude generated a dashboard more quickly and easily than mine with only one mistake in the initial output using the same data. It uses a geographical visualization of the data and statistics on the top to give insights into global GDP and top economies. Similar to my dashboard, the controls allow users to select a year, top countries, and a specific country. This dashboard offers more insight into different metrics as it offered GDP per capita as well, providing valuable context about economic development. However, the AI-generated visualization is mostly one color, making it hard to explore GDP component breakdowns compared to my colored bar and donut graphs. Claude's geographical focus and limited color differentiation make it harder to explore GDP breakdowns compared to my stacked bar approach. While faster to develop, my custom solution enables deeper comparative economic analysis.

